Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/11/2005
PI Name:	Ahlers, Guenter Ph.D.		
Project Title:	Boundary Effects on Transport Properties and Dynamic Finite-Size Scaling Near the Superfluid-Transition Line of 4He		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	guenter@physics.ucsb.edu	**Fax:**	FY 805-893-4170
PI Organization Type:	UNIVERSITY	**Phone:**	805-893-3795
Organization Name:	University of California		
PI Address 1:	Department of Physics		
PI Address 2:			
PI Web Page:	http://www.nls.physics.ucsb.edu/		
City:	Santa Barbara	**State:**	CA
Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/01/2003	**End Date:**	03/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Liu, Feng-Chuan (JPL)		
Grant/Contract No.:	NAG 3 2872		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	The task objective will be accomplished by measuring the thermal conductivity of liquid helium confined in cylindrical tubes of diameters from 1 to 100 microns. The objective of BEST is to determine the effects of boundaries on the thermal conductivity of liquid helium-4 near the superfluid transition The result of this work will provide the first test of scaling for transport properties near a continuous phase transition.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	A comparison between experimental results and theoretical prdictions will enhance our ability to calculate properties of complex systems using renormalization-group and Monte Carlo methods.
Task Progress:	We measured the thermal conductivity of liquid helium confined in micro-channel plates with a 1x10 micron cross section, and simultaneously of bulk helium. The bulk measurements served to provide the bulk lambda-point as a fixed point on the temperature scale. Analysis of the data is now under way.
Bibliography Type:	Description: (Last Updated: 02/06/2013)
Articles in Peer-reviewed Journals	Genio E, Murphy D, Ahlers G, Liu FC, Liu YM. "Scaling and universality of the thermal conductivity of liquid 4He near the superfluid transition and in restricted geometries." Advances in Space Research. 2005;35(1):87-91. http://dx.doi.org/10.1016/j.asr.2003.03.046 , Jan-2005
Articles in Peer-reviewed Journals	Laemmersahl C, Ahlers G, Ashby N, Barmatz M, Biermann PL, Dittus H, Dohm V, Duncan R, Gibble K, Lipa J, Lockerbie N, Mulders N, Salomon C. "Experiments in fundamental physics scheduled and in development for the ISS." General Relativity and Gravitation. 2004 Mar;36(3):615-49. http://dx.doi.org/10.1023/B:GERG.0000010734.62571.b4 , Mar-2004